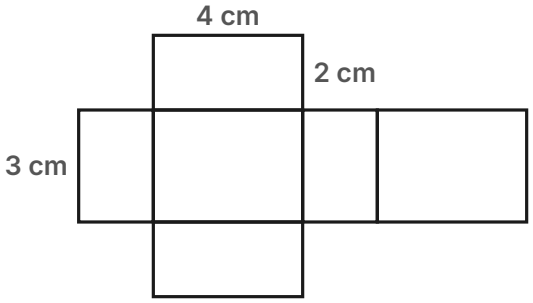
Surface Area and Nets

Open the box flat. Add the areas of all the faces.

A box 4 by 2 by 3, opened flat (a net)



6 faces, in 3 matching pairs.

top and bottom: 2 × 8 = 16

front and back: 2 × 12 = 24

two sides: 2 × 6 = 12

16 + 24 + 12 = 52 sq cm

Surface area is square units (covering). Volume is cubic units (filling).

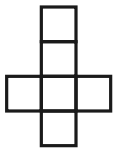
Your turn!

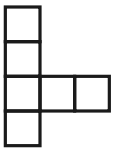
cube, edge 5 cm: 6 × ☐ = ☐ sq cm

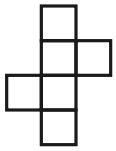
Surface Area and Nets: Practice

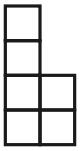
Find each face. Matching faces come in pairs.

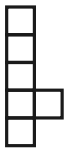
Part 1 Does it fold into a cube? Imagine folding it up. Circle yes or no.

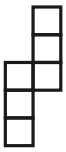
1)  yes no

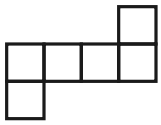
2)  yes no

3)  yes no

4)  yes no

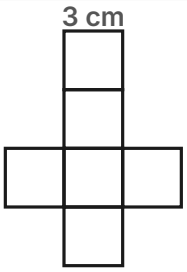
5)  yes no

6)  yes no

7)  yes no

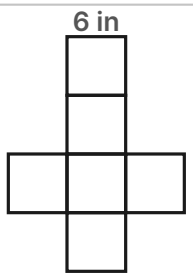
8)  yes no

Part 2 Cubes Six faces, all the same.

9) 

one face: ☐

6 × ☐ = ☐ sq cm

10) 

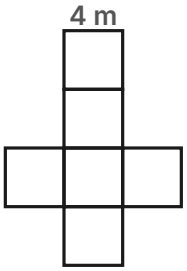
one face: ☐

6 × ☐ = ☐ sq in

Surface Area and Nets: Practice (continued)

Part 2 Cubes (continued)

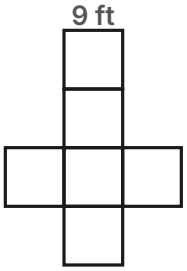
11)



one face: ☐

6 × ☐ = ☐ sq m

12)

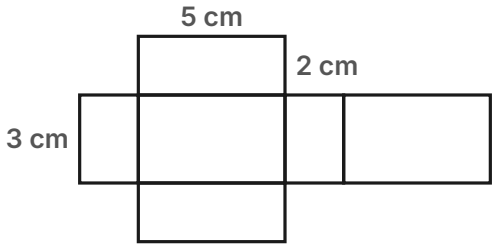


one face: ☐

6 × ☐ = ☐ sq ft

Part 3 Boxes Three pairs of faces.

13)



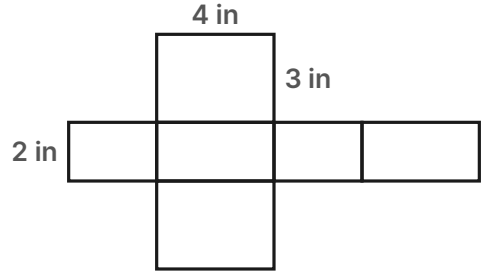
top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

two sides: 2 × ☐ = ☐

surface area = ☐ sq cm

14)



top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

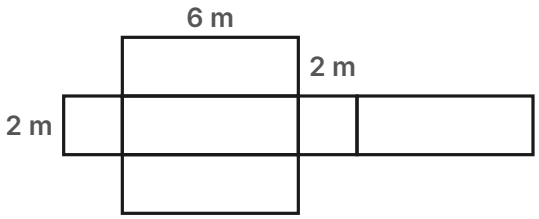
two sides: 2 × ☐ = ☐

surface area = ☐ sq in

Surface Area and Nets: Practice (continued)

Part 3 Boxes (continued)

15)



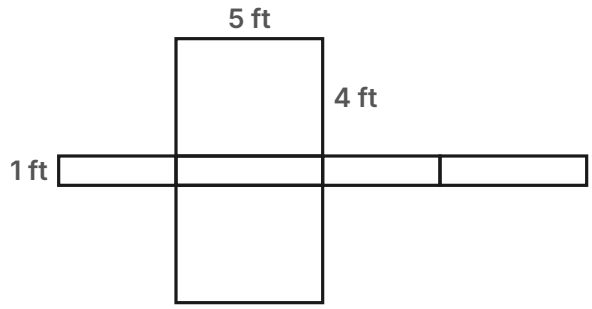
top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

two sides: 2 × ☐ = ☐

surface area = ☐ sq m

16)



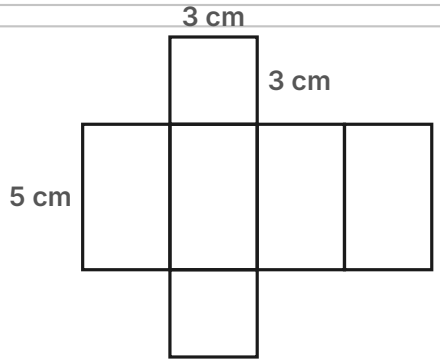
top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

two sides: 2 × ☐ = ☐

surface area = ☐ sq ft

17)



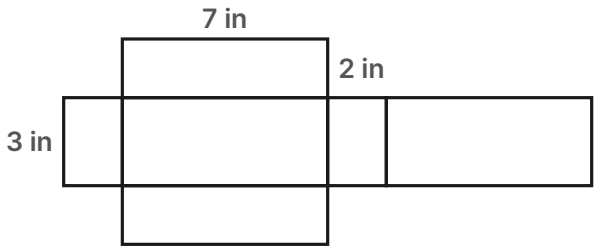
top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

two sides: 2 × ☐ = ☐

surface area = ☐ sq cm

18)



top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

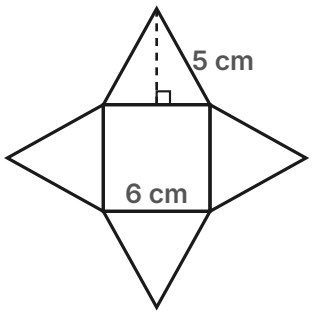
two sides: 2 × ☐ = ☐

surface area = ☐ sq in

Surface Area and Nets: Practice (continued)

Part 4 Square pyramids The square, then 4 triangles.

19)

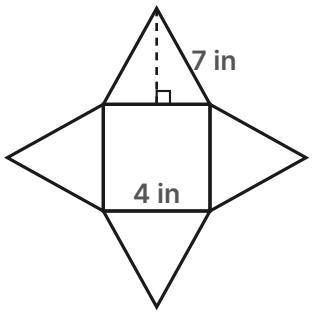


square: ☐

4 × ☐ = ☐

surface area = ☐ sq cm

20)

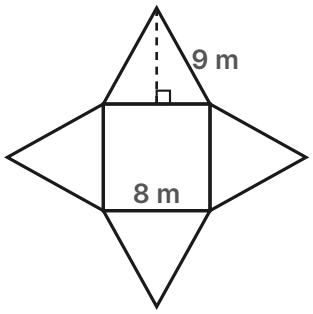


square: ☐

4 × ☐ = ☐

surface area = ☐ sq in

21)

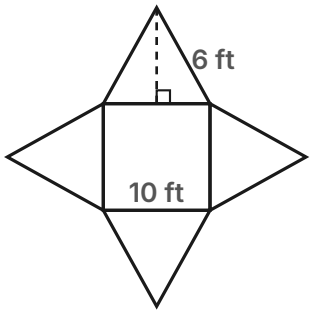


square: ☐

4 × ☐ = ☐

surface area = ☐ sq m

22)



square: ☐

4 × ☐ = ☐

surface area = ☐ sq ft

Surface Area and Nets: Practice (continued)

Part 5 Boxes No help this time.

23) box 6 × 4 × 3 cm: surface area = ☐ sq cm

24) box 5 × 5 × 2 in: surface area = ☐ sq in

25) box 8 × 3 × 2 m: surface area = ☐ sq m

26) box 7 × 4 × 5 ft: surface area = ☐ sq ft

27) box 10 × 2 × 3 cm: surface area = ☐ sq cm

28) box 9 × 6 × 1 m: surface area = ☐ sq m

29) box 4 × 4 × 7 in: surface area = ☐ sq in

30) box 12 × 3 × 2 cm: surface area = ☐ sq cm

Part 6 Cubes No help this time.

31) cube, edge 7 cm: surface area = ☐ sq cm

32) cube, edge 2 m: surface area = ☐ sq m

33) cube, edge 8 in: surface area = ☐ sq in

Surface Area and Nets: Practice (continued)

Part 6 Cubes (continued)

34) cube, edge 10 ft: surface area = ☐ sq ft

35) cube, edge 11 cm: surface area = ☐ sq cm

36) cube, edge 12 in: surface area = ☐ sq in

Part 7 More boxes

37) box 12 × 9 × 3 ft: surface area = ☐ sq ft

38) box 11 × 7 × 1 cm: surface area = ☐ sq cm

39) box 15 × 8 × 8 in: surface area = ☐ sq in

40) box 11 × 4 × 8 ft: surface area = ☐ sq ft

41) box 13 × 7 × 7 m: surface area = ☐ sq m

42) box 10 × 7 × 8 m: surface area = ☐ sq m

43) box 7 × 2 × 6 cm: surface area = ☐ sq cm

44) box 6 × 2 × 2 ft: surface area = ☐ sq ft

Surface Area and Nets: Practice (continued)

Part 7 More boxes (continued)

45) box 12 × 7 × 7 m: surface area = ☐ sq m

46) box 13 × 10 × 1 cm: surface area = ☐ sq cm

47) box 5 × 3 × 3 cm: surface area = ☐ sq cm

48) box 10 × 4 × 6 m: surface area = ☐ sq m

49) box 13 × 3 × 1 m: surface area = ☐ sq m

50) box 11 × 9 × 5 in: surface area = ☐ sq in

51) box 9 × 6 × 2 ft: surface area = ☐ sq ft

52) box 14 × 7 × 8 cm: surface area = ☐ sq cm

53) box 12 × 5 × 4 cm: surface area = ☐ sq cm

54) box 14 × 5 × 2 cm: surface area = ☐ sq cm

Surface Area and Nets: Practice (continued)

Part 8 More cubes

55) **cube, edge 8 ft: surface area =** ☐ **sq ft**

56) **cube, edge 12 cm: surface area =** ☐ **sq cm**

57) **cube, edge 17 cm: surface area =** ☐ **sq cm**

58) **cube, edge 4 ft: surface area =** ☐ **sq ft**

59) **cube, edge 19 m: surface area =** ☐ **sq m**

60) **cube, edge 20 ft: surface area =** ☐ **sq ft**

61) **cube, edge 14 m: surface area =** ☐ **sq m**

62) **cube, edge 3 ft: surface area =** ☐ **sq ft**

Surface Area and Nets: Practice (continued)

Part 9 Remember? Problems from earlier packets.

63) $2\frac{1}{2} \times 2 \times 6 =$ ☐

64) $5x = 85$: $x =$ ☐

65) $2.6 + 13.75 =$ ☐

66) -9 ☐ -4

67) when $n = 6$: $3n + 11 =$ ☐

68) 18 out of 20 = ☐ %

Part 10 Word problems Draw the net. Add the faces.

69) A cube box has edges of 4 in. How much paper covers it all?

Number sentence _____

Answer ☐ sq in

Draw or show your work

70) A box is 5 cm by 3 cm by 2 cm. What is its surface area?

Number sentence _____

Answer ☐ sq cm

Draw or show your work

71) A gift box is 10 in by 6 in by 4 in. How much wrapping paper covers it?

Number sentence _____

Answer ☐ sq in

Draw or show your work

Surface Area and Nets: Practice (continued)

Part 10 Word problems (continued)

72) A tent has 4 triangle sides, each with base 3 m and height 4 m. How much cloth for the sides?

Number sentence _____

Answer ☐ sq m

Draw or show your work

73) Challenge: A box 6 × 4 × 2 ft has no top. How much wood makes it?

Number sentence _____

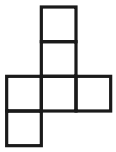
Answer ☐ sq ft

Draw or show your work

Check-In: Surface Area and Nets

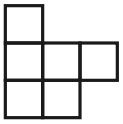
Show what you know. No help this time. Score: _____ / 6

Part 1 Does it fold into a cube? Circle yes or no.

1) 

yes

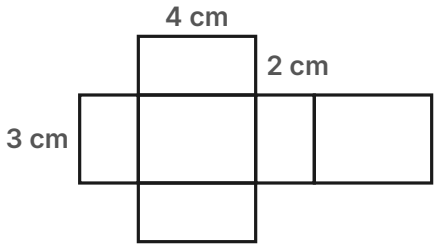
no

2) 

yes

no

Part 2 Boxes

3) 

top and bottom: 2 × ☐ = ☐

front and back: 2 × ☐ = ☐

two sides: 2 × ☐ = ☐

surface area = ☐ sq cm

Part 3 Find the surface area

4) **cube, edge 5 in: surface area = ☐ sq in**

5) **cube, edge 9 m: surface area = ☐ sq m**

6) **box 8 × 2 × 3 in: surface area = ☐ sq in**